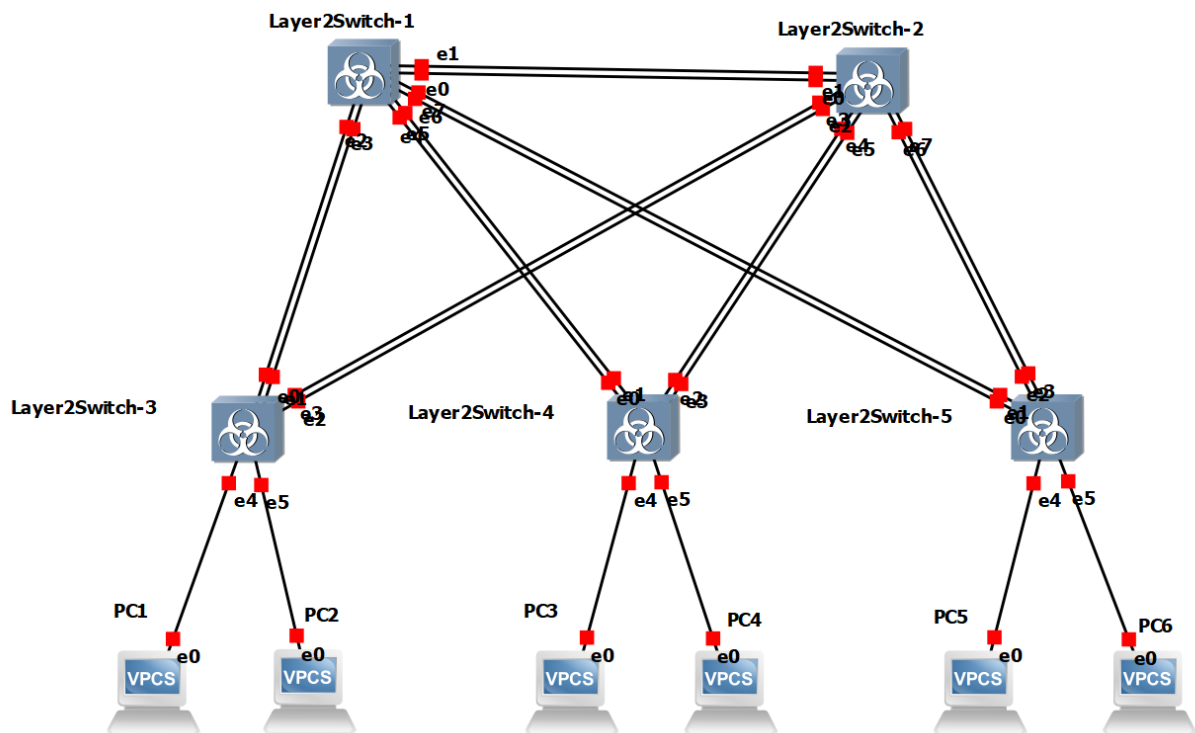




1) Для заданной на схеме сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

>show spa на всех коммутаторах

```

Layer2Switch-1 - PutTY
G11/3 Desg FWD 4 128.8 Shr
G12/0 Desg FWD 4 128.9 Shr

vIOS-L2-01>show spa
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0c01.6a73.0000
Cost 4
Port 5 (GigabitEthernet1/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0c1a.3c4f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
G10/0 Desg FWD 4 128.1 Shr
G10/1 Altn BLK 4 128.2 Shr
G10/2 Desg FWD 4 128.3 Shr
G10/3 Desg FWD 4 128.4 Shr
G11/0 Root FWD 4 128.5 Shr
G11/1 Altn BLK 4 128.6 Shr
G11/2 Desg FWD 4 128.7 Shr
G11/3 Desg FWD 4 128.8 Shr
G12/0 Desg FWD 4 128.9 Shr

vIOS-L2-01>

Layer2Switch-2 - PutTY
G12/0 Desg FWD 4 128.9 Shr

vIOS-L2-01>show spa
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0c01.6a73.0000
Cost 4
Port 5 (GigabitEthernet1/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0c15.5427.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
G10/0 Desg FWD 4 128.1 Shr
G10/1 Desg FWD 4 128.2 Shr
G10/2 Desg FWD 4 128.3 Shr
G10/3 Desg FWD 4 128.4 Shr
G11/0 Root FWD 4 128.5 Shr
G11/1 Altn BLK 4 128.6 Shr
G11/2 Desg FWD 4 128.7 Shr
G11/3 Desg FWD 4 128.8 Shr
G12/0 Desg FWD 4 128.9 Shr

vIOS-L2-01>

Layer2Switch-3 - PutTY
G11/3 Desg FWD 4 128.8 Shr
G12/0 Desg FWD 4 128.9 Shr

vIOS-L2-01>show spa
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0c01.6a73.0000
Cost 4
Port 3 (GigabitEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0c1b.3075.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
G10/0 Altn BLK 4 128.1 Shr
G10/1 Altn BLK 4 128.2 Shr
G10/2 Root FWD 4 128.3 Shr
G10/3 Altn BLK 4 128.4 Shr
G11/0 Desg FWD 4 128.5 Shr
G11/1 Desg FWD 4 128.6 Shr

vIOS-L2-01>

Layer2Switch-4 - PutTY
G10/3 Desg FWD 4 128.3 Shr
G11/0 Desg FWD 4 128.5 Shr
G11/1 Desg FWD 4 128.6 Shr

vIOS-L2-01>show spa
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0c01.6a73.0000
Cost 4
Port 5 (GigabitEthernet1/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0c01.6a73.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
G10/0 Desg FWD 4 128.1 Shr
G10/1 Desg FWD 4 128.2 Shr
G10/2 Desg FWD 4 128.3 Shr
G10/3 Desg FWD 4 128.4 Shr
G11/0 Root FWD 4 128.5 Shr
G11/1 Desg FWD 4 128.6 Shr

vIOS-L2-01>

Layer2Switch-5 - PutTY
G10/3 Altn BLK 4 128.4 Shr
G11/0 Desg FWD 4 128.5 Shr
G11/1 Desg FWD 4 128.6 Shr

vIOS-L2-01>show spa
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0c01.6a73.0000
Cost 4
Port 3 (GigabitEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0c03.d902.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
G10/0 Altn BLK 4 128.1 Shr
G10/1 Altn BLK 4 128.2 Shr
G10/2 Root FWD 4 128.3 Shr
G10/3 Altn BLK 4 128.4 Shr
G11/0 Desg FWD 4 128.5 Shr
G11/1 Desg FWD 4 128.6 Shr

vIOS-L2-01>

```

На первом коммутаторе:

>enable

#conf t

```
(config)# spa vlan 1 prio 32768
```

```
(config)# end
```

```
# write
```

```
LayerSwitch-1 - PuTTY
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 1
Address 0c1a.3c6f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)
Address 0c1a.3c6f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gig0/0 Desg FWD 4 128.1 Shr
Gig0/1 Desg FWD 4 128.2 Shr
Gig0/2 Desg FWD 4 128.3 Shr
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 1
Address 0c1a.3c6f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)
Address 0c1a.3c6f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gig0/0 Desg FWD 4 128.1 Shr
Gig0/1 Desg FWD 4 128.2 Shr
Gig0/2 Desg FWD 4 128.3 Shr
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

LayerSwitch-2 - PuTTY
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 1
Address 0c1a.3c6f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)
Address 0c1a.3c6f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gig0/0 Desg FWD 4 128.1 Shr
Gig0/1 Desg FWD 4 128.2 Shr
Gig0/2 Desg FWD 4 128.3 Shr
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

LayerSwitch-3 - PuTTY
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 1
Address 0c1a.3c6f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)
Address 0c1a.3c6f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gig0/0 Desg FWD 4 128.1 Shr
Gig0/1 Desg FWD 4 128.2 Shr
Gig0/2 Desg FWD 4 128.3 Shr
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

LayerSwitch-4 - PuTTY
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 1
Address 0c1a.3c6f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)
Address 0c1a.3c6f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gig0/0 Desg FWD 4 128.1 Shr
Gig0/1 Desg FWD 4 128.2 Shr
Gig0/2 Desg FWD 4 128.3 Shr
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

LayerSwitch-5 - PuTTY
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 1
Address 0c1a.3c6f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)
Address 0c1a.3c6f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gig0/0 Desg FWD 4 128.1 Shr
Gig0/1 Desg FWD 4 128.2 Shr
Gig0/2 Desg FWD 4 128.3 Shr
Gig0/3 Desg FWD 4 128.4 Shr
Gig0/0 Desg FWD 4 128.5 Shr
Gig1/1 Desg FWD 4 128.6 Shr
Gig1/2 Desg FWD 4 128.7 Shr
Gig1/3 Desg FWD 4 128.8 Shr
Gig1/0 Desg FWD 4 128.9 Shr
```

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

Каждому VPCs назначен ip-адрес от 192.0.0.1/24 до 192.0.0.6/24. С каждого VPC выполнено:

```
> ping 192.0.0.x
```

Где x – от 1 до 6, кроме собственного значения

```
VPC1 - PuTTY
Checking for duplicate address...
VPC1 : 192.0.0.2 255.255.255.0
VPC1> ping 192.0.0.1
84 bytes from 192.0.0.1 icmp_seq=1 ttl=64 time=11.311 ms
^C
VPC1> ping 192.0.0.2
84 bytes from 192.0.0.2 icmp_seq=1 ttl=64 time=9.468 ms
^C
VPC1> ping 192.0.0.3
84 bytes from 192.0.0.3 icmp_seq=1 ttl=64 time=10.686 ms
^C
VPC1> ping 192.0.0.4
84 bytes from 192.0.0.4 icmp_seq=1 ttl=64 time=10.686 ms
^C
VPC1> ping 192.0.0.5
84 bytes from 192.0.0.5 icmp_seq=1 ttl=64 time=6.032 ms
^C
VPC1> ping 192.0.0.6
84 bytes from 192.0.0.6 icmp_seq=1 ttl=64 time=13.477 ms
^C
VPC1>

VPC2 - PuTTY
Checking for duplicate address...
VPC2 : 192.0.0.3 255.255.255.0
VPC2> ping 192.0.0.1
84 bytes from 192.0.0.1 icmp_seq=1 ttl=64 time=9.462 ms
^C
VPC2> ping 192.0.0.2
84 bytes from 192.0.0.2 icmp_seq=1 ttl=64 time=9.553 ms
^C
VPC2> ping 192.0.0.4
84 bytes from 192.0.0.4 icmp_seq=1 ttl=64 time=9.700 ms
^C
VPC2> ping 192.0.0.5
84 bytes from 192.0.0.5 icmp_seq=1 ttl=64 time=12.152 ms
^C
VPC2> ping 192.0.0.6
84 bytes from 192.0.0.6 icmp_seq=1 ttl=64 time=10.836 ms
^C
VPC2>

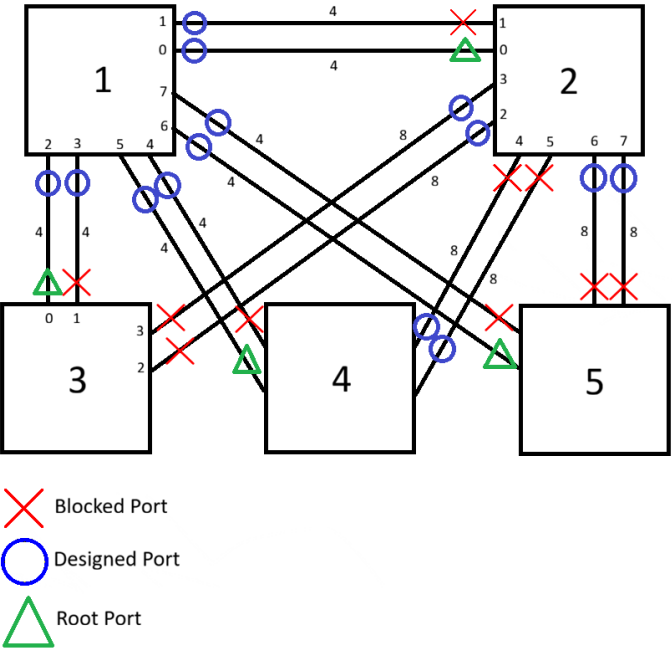
VPC3 - PuTTY
Checking for duplicate address...
VPC3 : 192.0.0.4 255.255.255.0
VPC3> ping 192.0.0.1
84 bytes from 192.0.0.1 icmp_seq=1 ttl=64 time=4.909 ms
^C
VPC3> ping 192.0.0.2
84 bytes from 192.0.0.2 icmp_seq=1 ttl=64 time=7.195 ms
^C
VPC3> ping 192.0.0.3
84 bytes from 192.0.0.3 icmp_seq=1 ttl=64 time=1.800 ms
^C
VPC3> ping 192.0.0.5
84 bytes from 192.0.0.5 icmp_seq=1 ttl=64 time=13.841 ms
^C
VPC3> ping 192.0.0.6
84 bytes from 192.0.0.6 icmp_seq=1 ttl=64 time=7.442 ms
^C
VPC3>

VPC4 - PuTTY
Checking for duplicate address...
VPC4 : 192.0.0.5 255.255.255.0
VPC4> ping 192.0.0.1
84 bytes from 192.0.0.1 icmp_seq=1 ttl=64 time=10.028 ms
^C
VPC4> ping 192.0.0.2
84 bytes from 192.0.0.2 icmp_seq=1 ttl=64 time=9.468 ms
^C
VPC4> ping 192.0.0.3
84 bytes from 192.0.0.3 icmp_seq=1 ttl=64 time=9.825 ms
^C
VPC4> ping 192.0.0.4
84 bytes from 192.0.0.4 icmp_seq=1 ttl=64 time=14.618 ms
^C
VPC4> ping 192.0.0.5
84 bytes from 192.0.0.5 icmp_seq=1 ttl=64 time=10.405 ms
^C
VPC4> ping 192.0.0.6
84 bytes from 192.0.0.6 icmp_seq=1 ttl=64 time=15.035 ms
^C
VPC4>

VPC5 - PuTTY
Checking for duplicate address...
VPC5 : 192.0.0.6 255.255.255.0
VPC5> ping 192.0.0.1
84 bytes from 192.0.0.1 icmp_seq=1 ttl=64 time=10.873 ms
^C
VPC5> ping 192.0.0.2
84 bytes from 192.0.0.2 icmp_seq=1 ttl=64 time=10.642 ms
^C
VPC5> ping 192.0.0.3
84 bytes from 192.0.0.3 icmp_seq=1 ttl=64 time=2.636 ms
^C
VPC5> ping 192.0.0.4
84 bytes from 192.0.0.4 icmp_seq=1 ttl=64 time=2.837 ms
^C
VPC5> ping 192.0.0.5
84 bytes from 192.0.0.5 icmp_seq=1 ttl=64 time=1.871 ms
^C
VPC5>

VPC6 - PuTTY
Checking for duplicate address...
VPC6 : 192.0.0.6 255.255.255.0
VPC6> ping 192.0.0.1
84 bytes from 192.0.0.1 icmp_seq=1 ttl=64 time=6.347 ms
^C
VPC6> ping 192.0.0.2
84 bytes from 192.0.0.2 icmp_seq=1 ttl=64 time=6.875 ms
^C
VPC6> ping 192.0.0.3
84 bytes from 192.0.0.3 icmp_seq=1 ttl=64 time=2.159 ms
^C
VPC6> ping 192.0.0.4
84 bytes from 192.0.0.4 icmp_seq=1 ttl=64 time=6.785 ms
^C
VPC6> ping 192.0.0.5
84 bytes from 192.0.0.5 icmp_seq=1 ttl=64 time=7.747 ms
^C
VPC6>
```

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл



4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

1-2

797	183.368039	0c:1a:3c:6f:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8002
798	184.146991	0c:15:54:27:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:15:54:27:00:00	Cost = 0	Port = 0x8002
799	184.150968	0c:15:54:27:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:15:54:27:00:00	Cost = 0	Port = 0x8002
800	184.159965	0c:15:54:27:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:15:54:27:00:00	Cost = 0	Port = 0x8002

1-3

323	74.062058	0c:1a:3c:6f:00:02	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8003
324	74.102883	0c:1a:3c:6f:00:02	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8003
325	74.111853	0c:1a:3c:6f:00:02	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8003
326	74.115881	0c:1a:3c:6f:00:02	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8003
328	74.604058	0c:18:30:79:00:00	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:18:30:79:00:00	Cost = 0	Port = 0x8001
329	74.606135	0c:18:30:79:00:00	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:18:30:79:00:00	Cost = 0	Port = 0x8001
330	74.608043	0c:18:30:79:00:00	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:18:30:79:00:00	Cost = 0	Port = 0x8001
331	75.065036	0c:1a:3c:6f:00:02	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8003

1-4

119	27.145266	0c:1a:3c:6f:00:04	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8005
121	28.000942	0c:01:6a:73:00:00	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:01:6a:73:00:00	Cost = 0	Port = 0x8001
122	28.003672	0c:01:6a:73:00:00	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:01:6a:73:00:00	Cost = 0	Port = 0x8001
123	28.007923	0c:01:6a:73:00:00	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:01:6a:73:00:00	Cost = 0	Port = 0x8001
124	28.145432	0c:1a:3c:6f:00:04	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8005
125	28.247331	0c:1a:3c:6f:00:04	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8005
126	28.256618	0c:1a:3c:6f:00:04	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8005
127	28.261309	0c:1a:3c:6f:00:04	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8005

1-5

65	15.012064	0c:1a:3c:6f:00:07	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8008
66	15.110060	0c:1a:3c:6f:00:07	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8008
67	15.118029	0c:1a:3c:6f:00:07	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8008
68	15.123051	0c:1a:3c:6f:00:07	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8008
69	16.012924	0c:1a:3c:6f:00:07	Nearest-Customer-Bridge STP	60	Conf. Root = 0/1/0c:1a:3c:6f:00:00	Cost = 0	Port = 0x8008
70	16.605597	0c:e3:d9:02:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/100/0c:e3:d9:02:00:00	Cost = 0	Port = 0x8002
71	16.607095	0c:e3:d9:02:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/200/0c:e3:d9:02:00:00	Cost = 0	Port = 0x8002
72	16.611170	0c:e3:d9:02:00:01	Nearest-Customer-Bridge STP	60	Conf. Root = 32768/300/0c:e3:d9:02:00:00	Cost = 0	Port = 0x8002

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

На втором коммутаторе:

```
> ena
```

```
# conf t
```

```
(config)# int gi0/0
```

```
(config-if)# spa cost 50
```

```
(config-if)# end
```

```
# write
```

